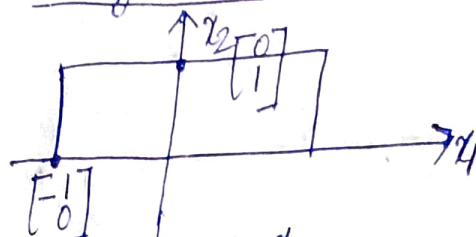


Transformation

Reflection through the x_2 -axis

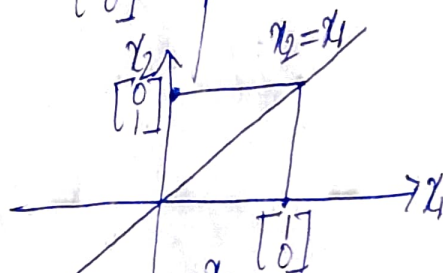
Image of the Unit Square



Standard Matrix

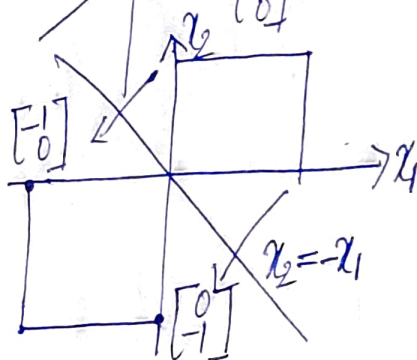
$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

Reflection through the line $x_2 = x_1$



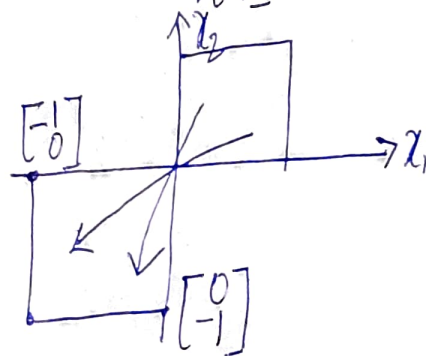
$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Reflection through the line $x_2 = -x_1$



$$\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

Reflection through the origin



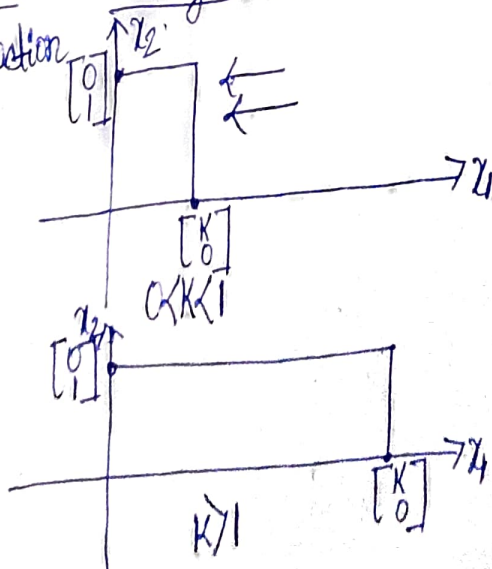
$$\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

Table 2 Contractions and Expansions

Transformation

Horizontal contraction and expansion

Image of the Unit Square



Standard Matrix

$$\begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}$$